FIITJEE

ALL INDIA TEST SERIES

FULL TEST – VII

JEE (Main)-2022

TEST DATE: 27-02-2022

Time Allotted: 3 Hours **Maximum Marks: 300**

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts.**
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A and Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B :** Do any five questions out of 10 Questions.

Section-A (01 – 20, 31 – 50, 61 – 80) *contains 60 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

Section-B (21 – 30, 51 – 60, 81 – 90) *contains 30 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer. There is no negative marking.*

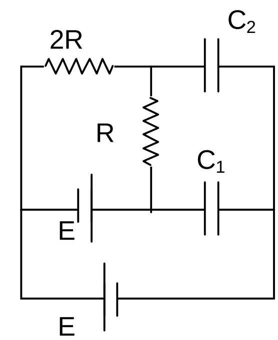
Physics

PART – A

SECTION – A
(One Options Correct Type)

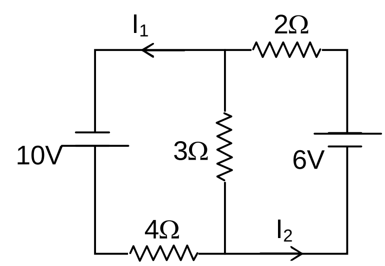
This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. If energy stored in the capacitor C_1 and C_2 are same, then the value of $\frac{C_2}{C_1}$ is



(A) $\frac{25}{36}$
(B) $\frac{36}{25}$
(C) $\frac{4}{9}$
(D) $\frac{5}{4}$

2. In the given circuit, find $I_2 - I_1$

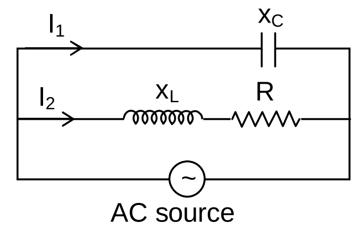


(A) zero
(B) $\frac{1}{13}$ A
(C) $\frac{2}{13}$ A
(D) $\frac{34}{13}$ A

3. A particle-1 of mass m, kinetic energy K and momentum P collides head on with another particle-2 of mass 2m at rest on horizontal smooth surface. If collision is elastic, find ratio of kinetic energy of particle -2 and particle-1 after collision.
(A) zero
(B) 1
(C) 1/8
(D) 8

4. Two blocks of mass m and 2m are connected by a spring of spring constant k. The system is placed on a horizontal frictionless surface. Now at t = 0, the block 2m is projected horizontally with velocity v_0 towards right. What is the minimum time when the elongation in spring is maximum?
(A) $\frac{\pi}{4}\sqrt{\frac{2m}{3k}}$
(B) $\frac{\pi}{2}\sqrt{\frac{2m}{3k}}$
(C) $\frac{3\pi}{4}\sqrt{\frac{2m}{3k}}$
(D) $\pi\sqrt{\frac{2m}{3k}}$

5. In the AC circuit drawn below find the phase difference between current I_1 and I_2.



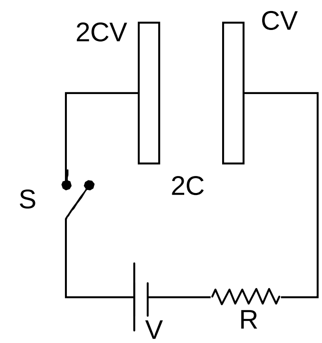
(A) $\frac{\pi}{2}+\tan^{-1}\left(\frac{X_L - X_C}{R}\right)$

(B) $\frac{\pi}{2}-\tan^{-1}\left(\frac{X_L - X_C}{R}\right)$

(C) $\frac{\pi}{2}+\tan^{-1}\left(\frac{X_L}{R}\right)$

(D) $\frac{\pi}{2}-\tan^{-1}\left(\frac{X_L}{R}\right)$

6. Plates of capacitor have charges 2CV and CV initially. Find the charge flowing through capacitor of capacitance 2C long time after switch is closed.



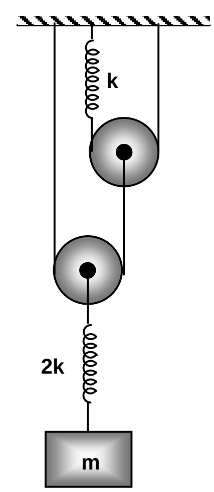
(A) zero

(B) $\frac{CV}{2}$

(C) CV

(D) $\frac{3CV}{2}$

7. In the figure drawn below, if pulleys are massless and frictionless. String is inextensible then find the time period of oscillation.



(A) $2\pi\sqrt{\frac{9m}{16k}}$

(B) $2\pi\sqrt{\frac{3m}{4k}}$

(C) $\pi\sqrt{\frac{9m}{16k}}$

(D) $\sqrt{\frac{3m}{4k}}$

8. A hydrogen atom is in an excited state of principal quantum number n. When it returns to the ground state, it emits a photon of wavelength λ_0. If R is Rydberg constant, then the value of n is

(A) $\sqrt{\frac{R\lambda_0 - 1}{R\lambda_0}}$

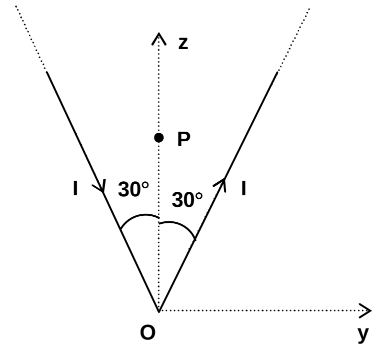
(B) $\sqrt{\frac{R\lambda_0 + 1}{R\lambda_0}}$

(C) $\sqrt{\frac{R\lambda_0}{R\lambda_0 - 1}}$

(D) $\sqrt{\frac{R\lambda_0}{R\lambda_0 + 1}}$

9. Two particles are projected horizontally and simultaneously from top of a tower in mutually perpendicular directions with same speed 30 m/s. After how much time their velocity vectors will be at angle 60° from each other (g = 10 m/s^2)
(A) 1 sec
(B) 3 sec
(C) 4 sec
(D) 5 sec

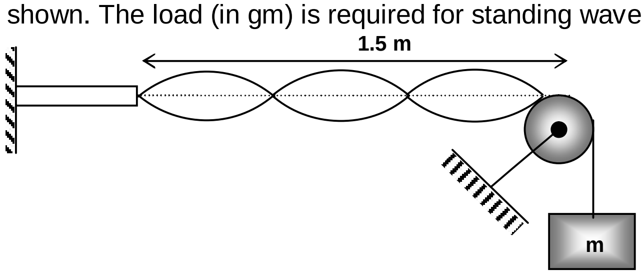
10. An infinite current carrying wire, carrying current I is bent in V shape, lying in y-z plane as shown in the figure. Find the magnetic field at P, if OP = 2a.



(A) $\frac{\mu_0 I}{2\pi a}(2+\sqrt{3})\hat{i}$
(B) $\frac{\mu_0 I}{2\pi a}(2-\sqrt{3})\hat{i}$
(C) $\frac{\mu_0 I}{4\pi a}(2+\sqrt{3})\hat{i}$
(D) $\frac{\mu_0 I}{4\pi a}(2-\sqrt{3})\hat{i}$

11. A certain mass of a gas undergoes a process given by dW = 2dU. If the molar heat capacity of gas for this process is 7.5R, then the gas is (where R is universal gas constant)
(A) Monoatomic gas
(B) Diatomic gas
(C) Polyatomic gas
(D) Insufficient data

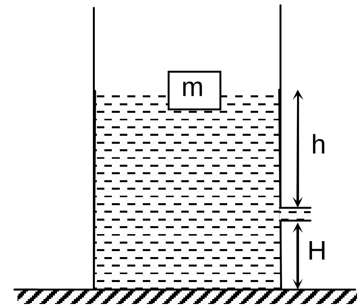
12. A string is stretched between pulley and a wave generator consisting of a plate vibrating up and down with small amplitude and frequency 120 Hz. The standing wave pattern has a 4 nodes as shown. The load (in gm) is required for standing wave with 5 nodes is (Here m = 256 gm initially)



(A) 216
(B) 512
(C) 288
(D) 144

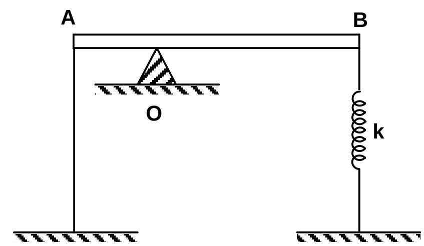
13. If a spring of stiffness k is cut into two parts A and B of length $\ell_A : \ell_B = 2:3$, then the stiffness of spring B is given by
(A) k
(B) $\frac{2k}{5}$
(C) $\frac{3k}{5}$
(D) $\frac{5k}{3}$

14. If the pin hole in the container is very small compared to the area of the base of container and a block floats in the ideal liquid then the speed of efflux is



(A) less than $\sqrt{2gh}$
(B) greater than $\sqrt{2gh}$
(C) equal to $\sqrt{2gh}$
(D) equal to $\sqrt{2gH}$

15. A uniform rod AB of mass 2kg and length 1m is placed on a sharp support O such that AO = 0.4m and OB = 0.6 m. A spring of force constant k = 600 N/m is attached to end B. To keep the rod horizontal, its end A is tied to a thread such that spring is elongated by 1 cm. The reaction of support on the rod when thread is burnt is

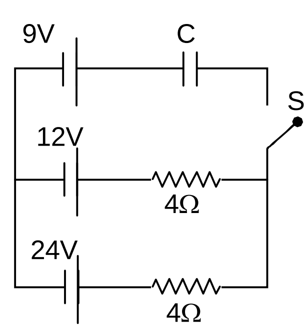


(A) 10 N
(B) 20 N
(C) 30 N
(D) 40 N

16. The electric field of two plane electromagnetic waves in vacuum are given by $\vec{E} = E_0 \cos(\omega t - kx)\hat{j}$ and $\vec{E}_2 = E_0 \cos(\omega t - ky)\hat{k}$. At t = 0, a particle of charge q is at origin with a velocity $\vec{v} = 0.8c\hat{j}$. The instantaneous force experienced by the particle is
(A) $qE_0(0.8\hat{i} + \hat{j} + 0.2\hat{k})$
(B) $qE_0(-0.8\hat{i} + \hat{j} + 0.2\hat{k})$
(C) $qE_0(0.8\hat{i} - \hat{j} + 0.2\hat{k})$
(D) $qE_0(-0.8\hat{i} - \hat{j} - 0.2\hat{k})$

17. 25 W/m^2 energy density of sunlight is normally incident on the surface of a solar panel. Some part of incident energy 25% is reflected from the surface and the rest 75% is absorbed what is the force exerted on 2 m^2 surface area of solar panel? (take c = 3 $\times$ 10^8 m/s)
(A) 10×10^{-8} N
(B) 20×10^{-8} N
(C) 25×10^{-8} N
(D) 35×10^{-8} N

18. In the given circuit, S is closed at t = 0. Find the power delivered by 24 V battery, when potential energy stored in capacitor is $\left(\frac{1}{4}\right)^{th}$ of its maximum value.



(A) 31.5 W
(B) 94.5 W
(C) 126 W
(D) 63 W

19. A beam of unpolarised light of intensity I_0 is passed through a polaroid A and then through another polaroid B which is oriented so that its principal plane makes an angle of 30° relative to that of A. The intensity of the emergent light is:

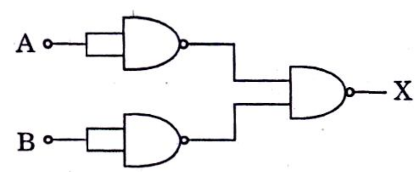
(A) $\frac{I_0}{2}$

(B) $\frac{3I_0}{8}$

(C) $\frac{I_0}{8}$

(D) I_0

20. The combination of gates shown below yields

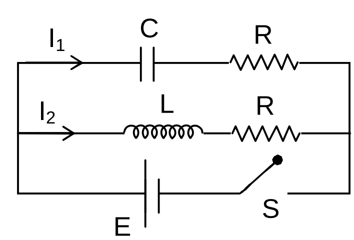


(A) XOR gate
(B) NAND gate
(C) NOT gate
(D) OR gate

SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

21. In the circuit drawn below initially capacitor C is uncharged and switch 'S' is closed at t = 0. If $C = \frac{1}{16}F$, L = 4H and R = 8Ω, the value of $\frac{I_2}{I_1}$ at t = $\ell n2$ sec will be

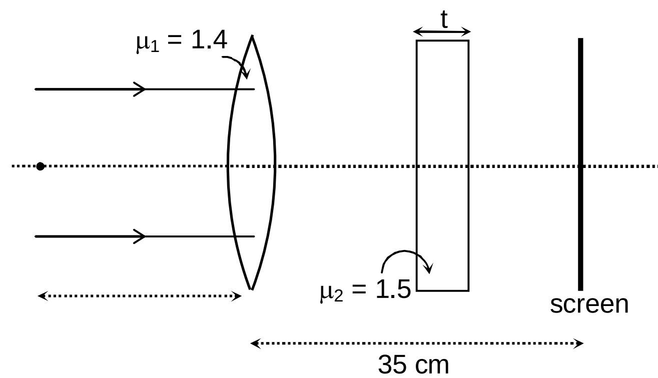


22. A soap bubble of radius R, surface tension T exists in equilibrium in vacuum. A monoatomic gas is trapped inside the soap bubble. Charge Q is given to bubble such that bubble expands from radius a to radius 2a under adiabatic condition. If surface charge on surface is found to be $\sqrt{\frac{x\varepsilon_0 T}{ya}}$. Find the value of x +y.

23. Two complete turns of the circular scale of a screw gauge covers a distance of 1 mm on its main scale. The number of divisions on the circular scale is 50. The screw gauge has a negative zero error of 0.03 mm. If the main scale reading is 3mm and circular scale reading is 35 during the measurement of diameter of wire, then find the diameter of wire in mm.

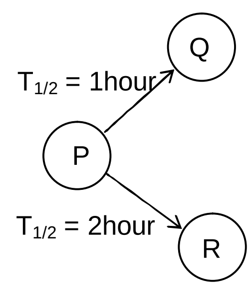
24. The output signal from an AM modulator is given as,
$S(t) = 5\cos(1800\,\pi t) + 20\cos(2000\pi t) + 5\cos(2200\,\pi t)$
What is the modulation index of the modulation wave?

25. A parallel beam is incident on a thin lens ($\mu_1 = 1.4$) and radius of curvature of each surface is 25 cm as shown in the figure. Find the thickness of slab in cm ($\mu_2 = 1.5$) between the lens and screen so that final image is formed on the screen if distance between lens and the screen is 35 cm.



26. A particle is executing SHM on a straight line. A and B are two points at which velocity is zero. It passes through a certain point P($P_A < P_B$) with a speed of $3\sqrt{2}$ m/s. It moves from A to P in time 0.5 sec and P to B in time 1.5 sec. Find the maximum speed in m/s.

27. A radioactive nuclei P decays simultaneously into two stable nuclei Q and R with half life of 1 hour and 2 hour respectively as shown in the figure. If the effective half life of P is $\frac{k}{9}$ hours, then find the value of k.



28. Two identical containers A and B with frictionless pistons contains the same ideal gas at the same temperature and the same volume V. The mass of the gas in A is m_A and that in B is m_B. The gas in each cylinder is now allowed to expand isothermally to the final volume 2V. The changes in the pressure in A and B are found to be ΔP and $1.5\Delta P$ respectively. Find $\frac{m_B}{m_A}$

29. Equation of standing wave in a string is given as,
$$y = 2\sin\left(2\pi x + \frac{\pi}{4}\right)\sin\left(200\pi t + \frac{\pi}{6}\right)\text{cm},$$
Where x is in cm and t is in second. Find the distance between two consecutive nodes in mm.

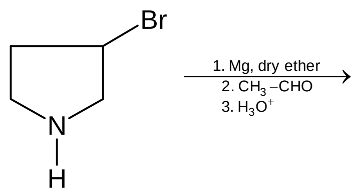
30. A galvanometer, whose resistance is 50 Ω, has 25 divisions on it. When a current of 4×10^{-4} A passes through it, its needle deflects by one division. What resistance in Ω should be connected to galvanometer to make it a voltmeter of range 2.5 V?

Chemistry

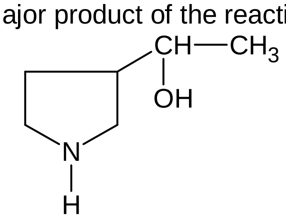
PART – B

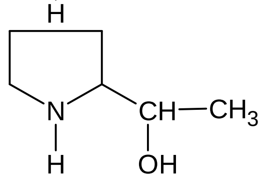
SECTION – A
(One Options Correct Type)

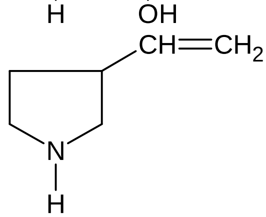
This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

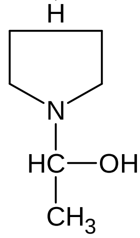
31. 

The major product of the reaction is

(A) 

(B) 

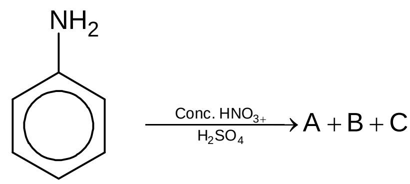
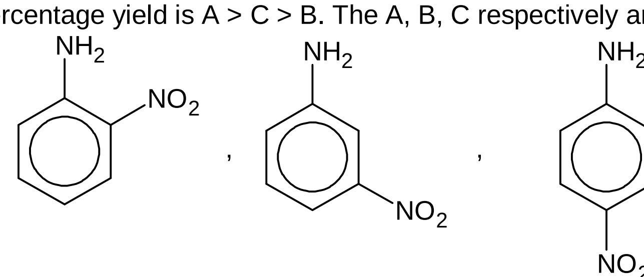
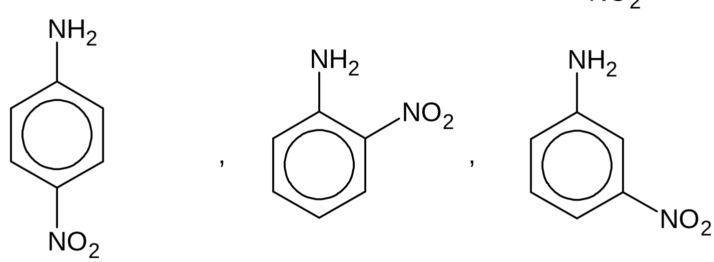
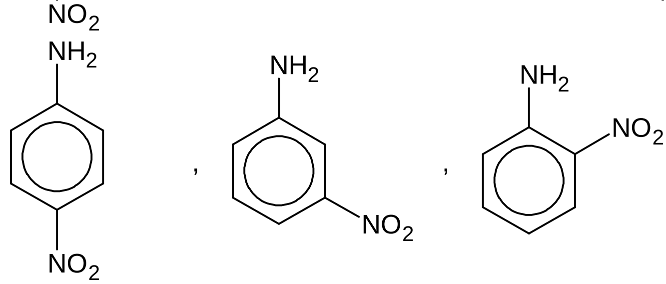
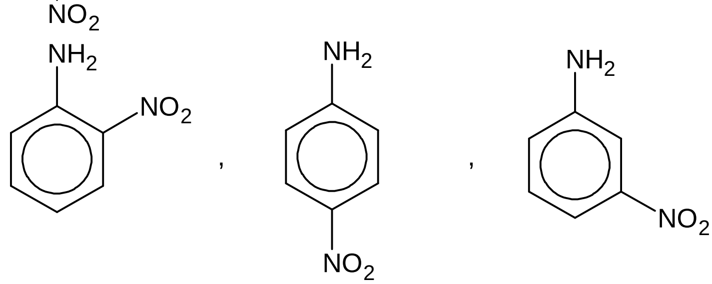
(C) 

(D) 

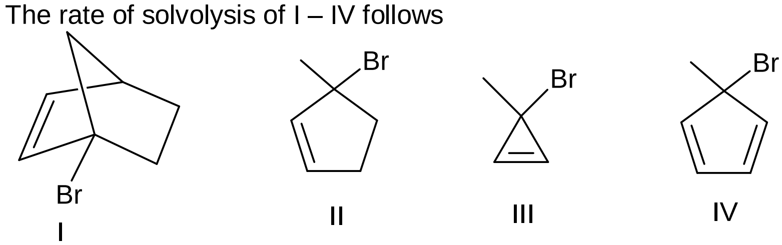
32. Myosin is the common example of
    (A) Vitamin
    (B) Fibrous proteins
    (C) Carbohydrates
    (D) Globular proteins

33. Which of the following is positively charged sol?
    (A) Copper sols
    (B) Sols of Gum
    (C) Oxides eg. TiO_2 sol
    (D) Metallic sulphides

34. Which of the following statements is incorrect?
(A) All Cu(II) halides are known except the iodide.
(B) The standard reduction electrode potential for Cu^{2+} to Cu has negative value.
(C) Permanganate titrations in presence of HCl are unsatisfactory as HCl is oxidized to chlorine.
(D) The E^o value for Mn^{3+}/Mn^{2+} couple is much more positive than that for Cr^{3+}/Cr^{2+} or Fe^{3+}/Fe^{2+}.

35.

The percentage yield is A > C > B. The A, B, C respectively are:
(A)

(B)

(C)

(D)


36. The rate of solvolysis of I – IV follows

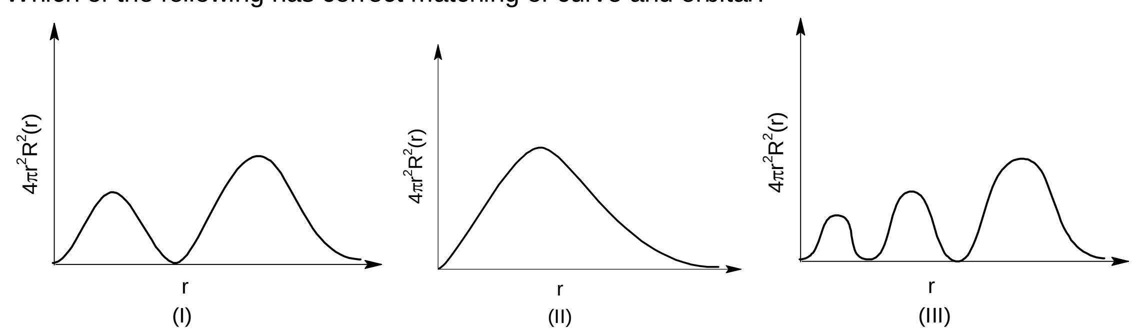


(A) I > II > III > IV
(B) III > I > II > IV
(C) III > II > I > IV
(D) IV > I > II > III

37. Which of the following precipitate of sulphide is yellow in colour?
(A) HgS
(B) PbS
(C) CuS
(D) CdS

38. Which of the following complex is a diamagnetic complex?
(A) $\left[Co\left(C_2O_4\right)_3\right]^{3-}$
(B) $\left[MnCl_6\right]^{3-}$
(C) $\left[Fe\left(CN\right)_6\right]^{3-}$
(D) $\left[CoF_6\right]^{3-}$

39. Which of the following has correct matching of curve and orbital?



(I) I(2p) II(1s) III(4p)
(II) I(3p) II(3d) III(3s)
(III) I(4d) II(2p) III(5d)
(IV) I(2s) II(4f) III(3d)
(A) II, III
(B) I, III
(C) III, IV
(D) All of the above

40. van der Waals equation for one mole of CO_2 gas at low pressure will be:

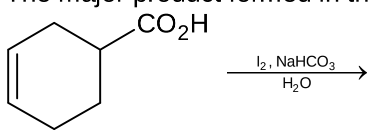
(A) $\left(P+\frac{a}{V^2}\right)V = RT$

(B) $P(V-b) = RT - \frac{a}{V^2}$

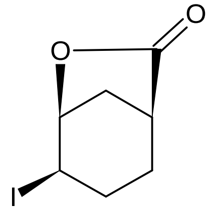
(C) $P = \frac{RT}{V-b}$

(D) $P = \left(\frac{RT}{V-b} - \frac{a}{V^2}\right)$

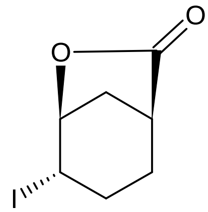
41. The major product formed in the following reaction is



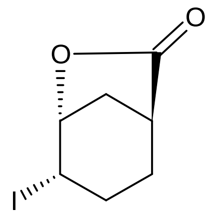
(A)



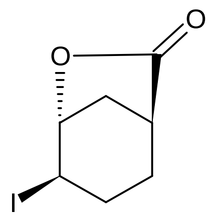
(B)



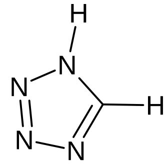
(C)



(D)



42. The correct statement for the following structures is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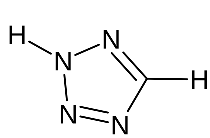
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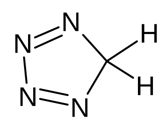

3

(A) 1, 2, 3 are resonance structures

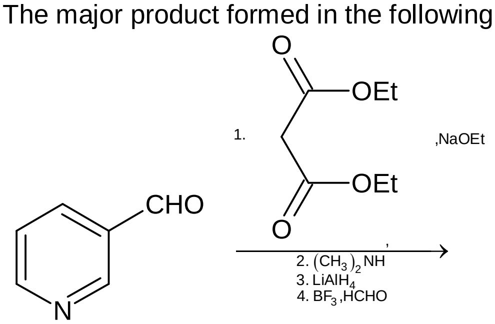
(B) 1 and 2 are resonance structures, where are 3 is an isomer of 1 and 2
(C) 1 and 3 are resonance structures, where as 2 is an isomer of 1 and 3
(D) 1, 2 and 3 are constitutional isomer

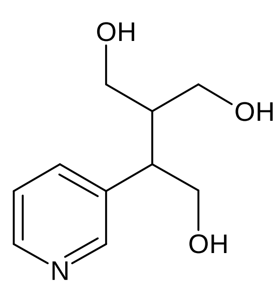
43. Which of the following compounds has a positive enthalpy of solution?
(A) LiF
(B) LiCl
(C) LiBr
(D) LiI

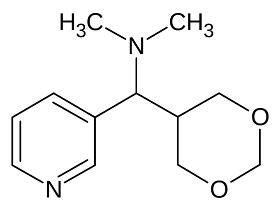
44. The temperature derivative of electrochemical cell potential E at constant pressure $\left(\frac{\partial E}{\partial T}\right)_P$ is given by
(A) $-\frac{\Delta S}{nF}$
(B) $\frac{\Delta S}{nF}$
(C) $\frac{\Delta S}{nFT}$
(D) $-\frac{\Delta S}{nFT}$

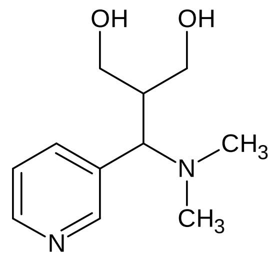
45. The d-orbitals involved in the hybridization to form square planar and trigonal bipyramidal geometries are respectively
(A) d_{z^2} and d_{z^2}
(B) d_{yz} and d_{z^2}
(C) $d_{x^2-y^2}$ and d_{z^2}
(D) $d_{x^2-y^2}$ and d_{yz}

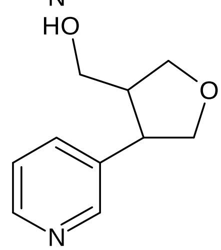
46. The major product formed in the following reaction sequence is:



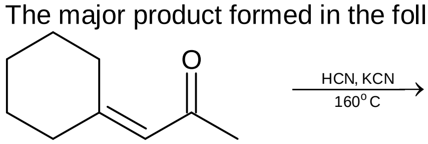
(A)


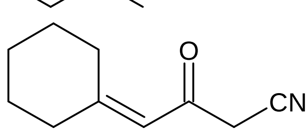
(B) 

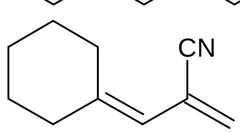
(C) 

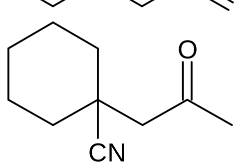
(D) 

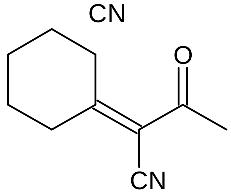
47. The major product formed in the following reaction is



(A) 

(B) 

(C) 

(D) 

48. The treatment of formic acid with concentrated sulphuric acid gives

(A) $HCHO + \frac{1}{2}O_2$

(B) $CO_2 + H_2$

(C) $CO + H_2O$

(D) $CO + H_2$

49. Zone refining may be employed for refining of metals like
(A) Ni and Zr
(B) Si and B
(C) Zn and Hg
(D) Cu and Ga

50. Which of the following order is correct?
(A) Li < Na < K < Rb < Cs (melting point)
(B) Li > Na > K > Rb > Cs (density)
(C) Be > Mg > Ca > Sr > Ba (ionization energy)
(D) Be < Mg < Ca < Sr < Ba (melting point)

SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

51. The maximum limit of nitrate in drinking water is________ppm.

52. Sea water containing 1 M NaCl has to be desalinated at 300 K using a membrane permeable only to water. The minimum pressure (in atm) required on sea water side of the membrane is______atm.

53. The dissociation constant of a weak monoprotic acid is 1.6×10^{-5} and its molar conductance at infinite dilution is 360.5×10^{-4} mho m^2 mol^{-1} for 0.01 M solution of this acid, the specific conductance is $n\times10^{-2}$ mho m^{-1}. The value of n is_________

54. A salt mixture (1 g) contains 25 wt% $MgSO_4$ and 75 wt% M_2SO_4. Aqueous solution of this salt mixture on treating with excess $BaCl_2$ solution results in the precipitation of 1.49 g of $BaSO_4$. The atomic mass of M in $g\ mol^{-1}$ is___________
(Atomic mass of Ba is 137 $g\ mol^{-1}$)

55. For the molecule $CH_3-CH=CH-CH(OH)-CH=CH-CH=C(CH_3)_2$ the number of all possible stereoisomers is

56. Calcium crystallises in fcc lattice of unit cell length $5.56\,\mathring{A}$ and density $1.4848\ g\ cm^{-3}$. The percentage of Schottky defects in the crystal is________

57. The standard reaction enthalpy of the reaction $Zn(s)+H_2O(g)\rightleftharpoons ZnO(s)+H_2(g)$ is $+223\ kJ\ mol^{-1}$ and the standard reaction Gibb's functions is $+33\ kJ\ mol^{-1}$ at 1520 K. Assuming that both ΔH^o and ΔS^o remains constant. The minimum temperature $x\times10^2$ K is the temperature above which equilibrium constant becomes greater than one. What is the value of x?

58. The rate of a first order reaction is $0.04\ mol\ litre^{-1}\ sec^{-1}$ at 10 minutes and $0.03\ mol\ litre^{-1}\ sec^{-1}$ at 22 minutes after initiation. The half-life of the reaction (in minutes) is

59. How many molecules of H_2O are required to hydrolyse 1 molecule of borazine?

60. One mole of an ideal gas (γ = 1.4) is expanded isothermally and reversibly at $27^{o}C$ till its volume is doubled. It is then adiabatically compressed reversibly to its original volume. The magnitude of total work done (in J) by the gas is__________

$\left(2^{0.4} = \frac{4}{3}, R = 8.3\ JK^{-1}, mol^{-1}\right)$

Mathematics PART – C

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. If $L_1 = \lim\limits_{x\to0} \dfrac{1-\cos x\cos 2x\cos 3x}{x^2}$ and $L_2 = \lim\limits_{x\to0} \dfrac{1-(\cos x)^{(\cos 2x)^{(\cos 3x)}}}{x^2}$, then the value of $|L_1 - L_2|$ is equal to
(A) 0
(B) $\dfrac{5}{2}$
(C) $\dfrac{13}{2}$
(D) $\dfrac{3}{2}$

62. If α is satisfying the equation $\sin^{-1}(e^x) - \sin^{-1}(x^2) = 0$, $\alpha \in R$ and β is number of solution of equation $\cos^{-1}\left(\dfrac{1-\tan^2\left(\frac{x}{2}\right)}{1+\tan^2\frac{x}{2}}\right) = |x-\pi| + \pi$, then which of the following is true?
(A) $\beta + \alpha > 0$
(B) $\beta + \alpha < 0$
(C) $\alpha > \beta$
(D) $\beta > 0$

63. In ΔABC if $a(2\cos A - 1) + b(2\cos B - 1) + c(2\cos C - 1) = 0$, then value of
$\dfrac{a^2\sin\left(\frac{A}{2}\right)}{bc} + \dfrac{b^2\sin\left(\frac{B}{2}\right)}{ac} + \dfrac{c^2\sin\left(\frac{C}{2}\right)}{ab}$ is
(A) 1
(B) 3
(C) $\dfrac{1}{2}$
(D) $\dfrac{3}{2}$

64. If $\alpha_1, \alpha_2, \alpha_3, \ldots, \alpha_9$ are the roots of the equation $\left(\dfrac{z+2}{z+1}\right)^9 = -1$, where $z = x + iy$, $x, y \in R$, $i = \sqrt{-1}$, then value of $\left|\sum\limits_{i=1}^{9}\alpha_i\right|$ is equal to
(A) $\dfrac{27}{2}$
(B) $\dfrac{9}{2}$
(C) 0
(D) 1

65. If matrix $M = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ and let M_1 and M_2 be two matrix such that $M_1 = M^4 - 4M^3 + 5M^2$, $M_2 = M^4 + 5M^2 - 2M$, then det(M_1M_2) is equal to
(A) 2^7
(B) 2^4
(C) 2^{13}
(D) 2^{11}

66. Let h(x) be a function such that h(x) = $\ln(1+x) - e^{\ln\left(\frac{x}{x+1}\right)}$, then which one is correct for h(x)?
(A) h(x) is decreases for some interval in its domain
(B) h(x) has two point of inflections
(C) h(x) is increasing in its domain
(D) $h'(x)$ is increasing for all x in its domain

67. The integral values of k for which the equation $x^2 + (2022 + k)x + 2022k + 1 = 0$ has integral roots, are k_1 and k_2, then focii of hyperbola $x^2 - y^2 = k_1 - k_2$ will be
(A) $\left(5\sqrt{2}, 0\right)$
(B) $\left(0, -6\sqrt{2}\right)$
(C) $\left(0, -2\sqrt{2}\right)$
(D) $\left(\sqrt{2}, 0\right)$

68. If a_1, a_2, a_3, ….., a_{120} are positive divisors of M, where $M = 2 \times 3^2 \times 5^3 \times 7^4$ and $P = \prod_{i=1}^{120} a_i$ & $Q = \frac{\sum_{i=1}^{120} a_i}{\sum_{i=1}^{120} \frac{1}{a_i}}$, then value of the following is true?
(A) $\frac{P}{Q} > M^{12}$
(B) $PQ < M^{12}$
(C) $\frac{P}{Q} < M^{12}$
(D) $PQ = M^{12}$

69. Let sides of a ΔPQR are represented y the equations PQ: $x + y + 6 = 0$, QR: $x - y - 2 = 0$ and RP: $y = 0$ and if the point $\left(t, -\sqrt{4 - t^2}\right)$ lying inside the triangle PQR, then number of integral value(s) of t is?
(A) 5
(B) 3
(C) 1
(D) 0

70. Let normals of an ellipse E at point A(–2, 3) and B(1, 2) intersect at M(0, 1) and S is the focus of the ellipse E, then length of SA is equal to
(A) $\frac{2}{\sqrt{10}}$
(B) $\frac{\sqrt{10}}{2}$
(C) $\frac{2}{3}$
(D) $\frac{3}{2}$

71. If a, b, c are positive real numbers such that $\frac{1}{a}+\frac{1}{b}+\frac{1}{c}=1$, then the minimum value of (a – 1)(b – 1)(c – 1) is
(A) 6
(B) 8
(C) 1
(D) 3

72. How many three digit numbers of the type $a_1a_2a_3$ (decimal representation) where $a_i \in \{0, 1, 2, \ldots, 9\}$, $a_1 \neq 0$ are formed such that $a_1 < a_2 > a_3$ (a_1, a_2, a_3 are distinct digits)
(A) 120
(B) 204
(C) 320
(D) none of these

73. If $\vec{a}, \vec{b}, \vec{c}$ are three vectors such that $|\vec{a}| = |\vec{b}| = \sqrt{3}$, $|\vec{c}| = \sqrt{6}$, $\vec{a}\cdot\vec{b} = 1$, $\vec{a}\cdot\vec{c} = 2$ and $\vec{b}\cdot\vec{c} = -2$, then the volume of parallelepiped formed by the vectors $\vec{a}, \vec{b}, \vec{c}$
(A) 4
(B) 16
(C) 2
(D) $\sqrt{12}$

74. Let $f(\theta) = \sqrt{16\cos^2\theta + 25\sin^2\theta} + \sqrt{25\cos^2\theta + 16\sin^2\theta}$ where $\theta \in R$, then value of $\frac{\max f(\theta)}{\min f(\theta)}$ is
(A) $\frac{\sqrt{41}}{5}$
(B) $\frac{\sqrt{41}}{9}$
(C) $\frac{\sqrt{82}}{5}$
(D) $\frac{\sqrt{82}}{9}$

75. If $\vec{p}, \vec{q}$ are non parallel unit vectors and $\vec{r}$ is a vector in a plane perpendicular to $\vec{p}$ and $\vec{q}$ such that $|\vec{r}| = 5$ and $|\vec{p} \times \vec{q}| = |\vec{p} + \vec{q}|^2$, then $\left|\left[\vec{p}\ \ \vec{q}\ \ \vec{r}\right]\right| + |\vec{p} \cdot \vec{q}|$ is equal to

(A) $\frac{23}{5}$

(B) $\frac{17}{5}$

(C) $\frac{3}{5}$

(D) $\frac{19}{5}$

76. Let the set of complex numbers (p_1, q_1), (p_2, q_2), (p_3, q_3), ... denoting the points on the complex plane satisfying the relation $(p_{n+1}, q_{n+1}) = \left(\sqrt{3}\, p_n - q_n, \sqrt{3}\, q_n + p_n\right)$ $\forall$ n = 1, 2, 3, ... if $(p_{100}, q_{100}) = (2, 4)$, then the value of $p_1 + q_1$ is

(A) $\frac{1}{2^{56}}$

(B) $\frac{1}{2^{97}}$

(C) $\frac{1}{2^{98}}$

(D) $\frac{1}{2^{99}}$

77. The number of real values of x, $x \in R$ satisfies the equation
$\sqrt{3x^2 - 18x + 52} + \sqrt{2x^2 - 12x + 162} = \sqrt{-x^2 + 6x + 280}$, is

(A) 1
(B) 2
(C) 3
(D) infinite

78. If standard deviation of the variate x is 5, then standard deviation of the variate $\frac{15x + 2022}{7}$ is

(A) $\frac{2022}{5}$

(B) $\frac{2077}{7}$

(C) $\frac{35}{15}$

(D) $\frac{75}{7}$

79. If y = g(x) is a function, g : [0, 3] $\to$ R, $g(x) = \tan^{-1}\left(\frac{\sqrt{3}x - 3x}{3\sqrt{3} + x^2}\right) + \tan^{-1}\left(\frac{x}{\sqrt{3}}\right)$, then the maximum value of g(x) is

(A) $\frac{\pi}{4}$
(B) $\frac{\pi}{3}$
(C) $\frac{3\pi}{4}$
(D) $\frac{\pi}{6}$

80. If there are n number of integers between 1 and 10000 which have the property that "sum of digits of the number is equal to 12", then n is
(A) 415
(B) 455
(C) 514
(D) 554

SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

81. If OAB is a triangle such that $\overrightarrow{OA} = \vec{a}$, $\overrightarrow{OB} = \vec{b}$, O is origin if D, E, F are points on the OB, AB, OA respectively such that $\overrightarrow{OF} = \frac{1}{3}\vec{a}$, $\overrightarrow{OD} = \frac{2}{3}\vec{b}$, $\overrightarrow{OE} = \frac{2\vec{a} + \vec{b}}{3}$ if AD, FB, OE intersect at R, Q, P points respectively, if $\frac{\text{area of } \Delta ABC}{\text{area of } \Delta PQR} = \frac{m}{n}$, m, n are coprime, then (m + n) equal to

82. How many three digit numbers abc (decimal representation) a $\neq$ 0 can be formed by using decimal digit with the property that a < b and c < b

83. If number of values of θ_1 for which tan $3\theta_1$ – tan $2\theta_1$ = 1 + tan $3\theta_1$ tan $2\theta_1$, $\theta_1 \in [0, 2022\pi]$ are λ_1 and number of value of θ_2 for which $\sin(\theta_2)^8 - \cos(\theta_2)^8 = 1$, $\theta_2 \in [0, 100\pi]$ are λ_2, then $\lambda_1 + \lambda_2$ is equal to

84. If z_1, z_2, z_3, z_4, z_5 are the roots of the equation $z^5 + z^4 + z^3 + z^2 + z + 1 = 0$ and if $\lambda = \sum_{1 \le i \le j \le 5} |z_i - z_j|^2$, then the value of λ is

85. There is a non-zero complex number z_3 such that $z_1^2 = z_2 z_3$ and $z_2^2 = z_1 z_3$, if $|z_1| = |z_2| = |z_3| = 6$ and if area of triangle formed by z_1, z_2 and z_3 as vertices are Δ, then Δ is equal to

86. In a triangular park ABC a pole of height h standing on the vertex A and there are three equidistant points on the side BC whose angle of depression are observed to be $\frac{\pi}{6}$, $\frac{\pi}{4}$ and $\frac{\pi}{3}$ respectively, then the value of h^2 is

87. If M be a square matrix of order 3×3 such that det(M) = 4, then det(adj (2M)) is equal to

88. If length of the path traced by the locus of mirror image of foci (both the focus) of ellipse $16x^2 + 4y^2 = 1$ with respect to its variable tangent $y = mx + c$ is L, then [L] is equal to (where [.] denotes greatest integer function)

89. Let AB is a focal chord and MN is latus rectum of the parabola $y^2 = 2x$, if AB makes angle $\frac{\pi}{6}$ with positive x-axis (point A and M in first quadrant), then the value of $\left(\frac{1}{\text{area of } \Delta\, \text{MAS}} + \frac{1}{\text{area of } \Delta\, \text{NBS}}\right)^2$, (where S is the focus of given parabola) is

90. If the sum of series $\sum_{p=1}^{3}\sum_{r=1}^{p}\sum_{n=1}^{20}\sum_{m=1}^{20}\tan^{-1}\left(\frac{m}{n}\right) = \lambda$, then the value of $\frac{\lambda}{\pi}$ is